

Transmission Expansion and Reliability in Peru's Energy Transition: A System Dynamics Approach

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Abstract

Peru combines abundant renewable resources with a generation mix still built on hydropower and natural gas, a longitudinal grid linking three load zones in series, and generation and transmission projects that advance on very different timescales. Which of these uncertainties governs the system's trajectory determines where planning effort should go. We build a system dynamics model of the Peruvian power system, coupled to a congestion-aware dispatch layer, and simulate it monthly from 2025 to 2050 over a designed experiment of 129 runs: a central case, eight one-at-a-time sensitivity runs, and 120 Latin-hypercube Monte Carlo samples of four uncertain drivers — fuel prices, demand growth, and project lead times on the generation and the transmission side. Demand growth dominates every outcome, with Spearman rank correlations of 0.99 for transmission length, 0.97 for installed capacity and -0.75 for the average price, while the other three drivers stay below 0.27 in absolute value. A $\pm 20\%$ band on the 2050 demand level moves required transmission by -18.9% to $+23.8\%$ and installed capacity by -24.0% to $+29.3\%$. The price moves against intuition: faster demand growth lowers the long-run average price, pulling forward low-cost renewable capacity the central case defers. The least-cost trajectory is consequently the most transmission-intensive, requiring 26 % more network. For Peru the binding planning question is therefore not how to hedge fuel-price risk but whether the grid can be delivered fast enough to serve demand growth that is itself the dominant unknown.

Keywords: Power system expansion planning, System dynamics, Transmission expansion, Uncertainty analysis, Latin hypercube sampling, Peru

1. Introduction

Expansion planning rests on numbers nobody knows. What fuel will cost a decade out, how fast demand will grow, and how late projects will actually come online are all uncertain, and each is routinely handled by picking a central value and running a scenario around it. That practice is defensible only if the uncertainties are of comparable weight. When one dominates, effort spent hedging the others is misallocated, and the planning question changes from *what might happen* to *which unknown is worth managing*.

Peru is a useful place to ask that question. Its generation mix is still built on hydropower and natural gas, with non-conventional renewables at roughly 5 % of generation despite a resource base that modelling work repeatedly finds economically attractive [14, 10]. Its transmission network is longitudinal: three load zones strung in series, with the best wind and solar resources systematically distant from the demand they would serve [46]. And its institutional record is mixed in an instructive way — the 1992 restructuring produced

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genuine competition and improved security of supply [34, 30], and Peru pioneered technology-specific renewable auctions, but that mechanism has since been suspended and investment in both generation and transmission slowed once privatisation ran its course. The country holds, simultaneously, a strong resource endowment, a grid not built for it, and a planning process whose delivery times are themselves uncertain.

Most of the modelling literature addresses this with optimisation. Generation and transmission expansion planning, jointly or separately, are mature fields [18, 4, 20, 44, 47]. These models are indispensable for identifying what an efficient expansion path would look like, but they answer a normative question under an assumption of perfect rationality and a single optimising agent. The system they describe is not the system that exists: in a liberalised market, generation investment is decentralised and driven by expected profitability, transmission is planned centrally on a longer clock, and the gap between the two is bridged by neither [12, 21]. It is in that gap that delays, congestion and stranded capacity appear.

System dynamics was developed for exactly this situation — feedback, accumulation and delay [17, 42] — and has a long record of reproducing the investment cycles and capacity gaps that optimisation abstracts away [16, 2, 3, 50]. What it is less often asked to do is to *quantify*, rather than illustrate, the relative weight of the uncertainties driving it. A single simulation, or a handful of named scenarios, shows what the model does; it does not show which input the answer depends on.

This paper closes that gap for Peru. We build a system dynamics model of the Peruvian system, coupled to a congestion-aware dispatch layer, with an explicit regulatory and construction pipeline for both generation and transmission. We subject it to a designed experiment of 129 simulations over 2025–2050: a central case, eight one-at-a-time runs isolating each driver, and 120 Latin-hypercube Monte Carlo samples in which four drivers move at once — fuel prices, demand growth, and project lead times on the generation and the transmission side.

The contribution is threefold: a **quantified ranking** of the uncertainties rather than an illustration of them, reported as one-at-a-time sensitivities and rank correlations with a convergence check; a **transmission-explicit representation** of a longitudinal system, in which grid built for demand, for generation connection and for interconnection each carry their own trigger and lead time, only the third being constrained by the series topology; and a **reproducible artefact**, with model, data, design, code and a notebook regenerating every figure openly available and the validation battery and full parameter set as supplementary material.

The central result bears directly on policy and is counter-intuitive. Higher demand growth does not raise the long-run average price in this system; it lowers it, by pulling forward low-cost renewable capacity that the central case defers. The least-cost trajectory is consequently the most transmission-intensive, requiring about a quarter more network. For a system whose grid is already the binding constraint, that reframes the planning problem: the risk to manage is not fuel-price exposure but whether transmission can be delivered fast enough to serve demand growth that is itself the dominant unknown.

2. Literature review

2.1. Transmission planning under the energy transition

Transmission is where the energy transition is most visibly constrained. The resources that decarbonise a system sit where the wind blows and the sun shines, not where load is concentrated, so every increment of renewable capacity implies an increment of network — and the network is planned centrally, built slowly, and remunerated under rules written for a different fleet.

The modelling response has largely been optimisation. Transmission expansion planning (TEP) treats a central planner maximising welfare subject to network constraints, extended steadily to accommodate the transition: stochastic formulations that make renewable hosting capacity an explicit objective [47] and

multi-stage investment games for offshore grids [44]. Joint generation and transmission formulations go further, and the coordination gain is substantial: Herding et al. [20] find that coordinating the two reduces total investment by around 10 % and network expansion by up to 30 %. Reviews of the field [18, 4] document both its maturity and its central assumption — a single rational agent optimising a well-posed objective.

That assumption is the difficulty here. In a liberalised market the generation investor and the network planner are different agents acting on different information over different horizons, and optimisation struggles to represent the structural and behavioural change policy induces [12]. Del Río et al. [11] make the point concretely for Latin America: the cost of capital, not resource quality, determines which projects are financed.

2.2. Infrastructure delays and reliability

A transmission project that is planned but late is, for the duration of the delay, indistinguishable from one that was never planned. Ritter et al. [33] simulate delayed interconnector expansion in a high-renewable European system and show the delay does not merely postpone benefits: it raises emissions and generation costs, because surplus renewable energy that cannot be moved is replaced by conventional generation elsewhere, altering the final mix. Herrera et al. [21] reach a structurally similar conclusion for northeast Brazil, where transmission constraints bound wind expansion directly. The mechanism is the same in both: generation investment responds to a connection opportunity the network has not yet delivered.

The causes of delay are largely non-technical. Right-of-way disputes, licensing and local opposition recur across jurisdictions; Sun et al. [43] quantify the last of these, and Boateng et al. [6] generalise the point, identifying institutional-quality thresholds below which renewable deployment responds very differently to identical incentives. For a system such as Peru's, lead-time risk is therefore better modelled as an institutional variable than an engineering one.

2.3. Renewable integration in emerging economies

Emerging economies face the integration problem with more expensive capital, thinner institutions, and grids that are radial or longitudinal rather than meshed. da Silva and Carneiro [39] find the regional efficiency of Brazilian renewable expansion varying sharply by region, consistent with network access rather than resource quality being the differentiator, while Filippo Antonioli et al. [15], Wu and Zhong [49] and Sheng et al. [38] document rooftop and community-scale generation as mechanisms that reduce the inter-zonal transfers a longitudinal grid must support.

The alternative to building network is using it harder. Grid-enhancing technologies raise transfer capability without new right-of-way — Sadeghian and Mohammadi-Ivatloo [35] formalise dynamic thermal line rating under chance constraints — though remuneration frameworks for them are generally absent. Storage plays the complementary role: Sousa et al. [41] quantify the substitution between storage and other flexibility resources, making the structural point that the transmission a system needs is not independent of how much flexibility it builds.

2.4. System dynamics in energy planning

System dynamics takes the opposite stance: rather than solving for the best configuration, it simulates the configuration the system's own feedback, accumulation and delay will produce [17, 42]. For electricity this is productive because the pathologies of interest — investment cycles, capacity gaps, price spikes following construction lags — are consequences of structure rather than of optimisation failure.

The lineage is long: Ford [16] explained the California capacity crisis as a construction-delay phenomenon, Arango [2] simulated alternative regulation in Colombia, Assili et al. [3] examined capacity

payments with construction delays explicit, and Simshauser [40] connected resource adequacy to investment uncertainty. Recent work extends this to transition pathways [50, 26, 24], carbon border adjustment [23] and PV adoption policy [29].

Model confidence in system dynamics is established by a documented battery of tests rather than a goodness-of-fit statistic [5, 28], and recent methodological work reconsiders what confidence means for models used exploratorily rather than predictively [32]. We follow that protocol explicitly and report it in full in the supplementary material.

Where this literature is thinnest is uncertainty quantification. Scenario analysis is near-universal; designed sampling of the input space is rare. The tools are standard outside system dynamics — Latin hypercube sampling [25] and regression- or variance-based global sensitivity measures [22] — and are used in power system work, for instance in Álvarez-Piñero et al.’s [2026] probabilistic assessment of the Spanish 2030 system. Combining them with a feedback-rich simulation model is the step this paper takes.

2.5. Peruvian and Latin American studies, and the gap

Peru’s restructuring is studied as a reference case: a reform that delivered competition and improved security of supply while accumulating a high reserve margin and fossil dependence [34, 30]. Recent work addresses the transition itself [14, 10] and the network [46]. Regionally, Sauma and Oren [36] established the proactive-versus-reactive distinction in transmission valuation, while analyses of Andean interconnection identify regulatory sovereignty rather than transfer capability as the barrier [19, 37].

Three gaps follow. First, the delay literature establishes that lead times matter but not how much, relative to other uncertainties, in a common design. Second, the system dynamics energy literature rarely quantifies uncertainty with a designed experiment, so a small number of named scenarios cannot separate a structural result from an artefact of one input assumption. Third, Peru is under-represented: existing studies are either optimisation-based assessments of a renewable target [14, 10] or operational analyses of the network [46], and none represents the investment-and-delay feedback between generation and transmission that a longitudinal, gas-dependent system with suspended auctions would be expected to exhibit. This paper addresses all three.

3. The Peruvian power system

3.1. Market structure and generation mix

Following a macroeconomic collapse in the late 1980s, Peru moved from an integrated state system to a competitive wholesale market with marginal nodal prices, capacity payments and bilateral contracts. The Electric Concessions Law of 1992 unbundled generation, transmission and distribution, and the supply auctions introduced from 2006 consolidated productivity gains, with clear efficiency improvements in distribution after restructuring [34, 30].

The generation matrix remains dominated by hydropower and natural gas. By 2015 the mix was roughly 50 % hydro and 45 % gas, with thermal sources overall accounting for about 41 % of generation by 2017 [34]. The dependence is structural: over half the natural gas consumed in the country goes to power plants [13], making electricity the primary driver of gas demand and exposing it to price and supply shocks. Despite substantial solar, wind, geothermal and biomass potential, non-conventional renewables supplied only about 5 % of generation by 2022 [8].

Fiestas-Chevez et al. [14] find that raising non-conventional renewables to 28 % of the matrix would cut gas consumption by 2.04 km³ and avoid 4.99 Mt CO₂-eq annually while reducing system costs, and De la Cruz Torres et al. [10] reach compatible conclusions to 2050. The economic case is established; the route is

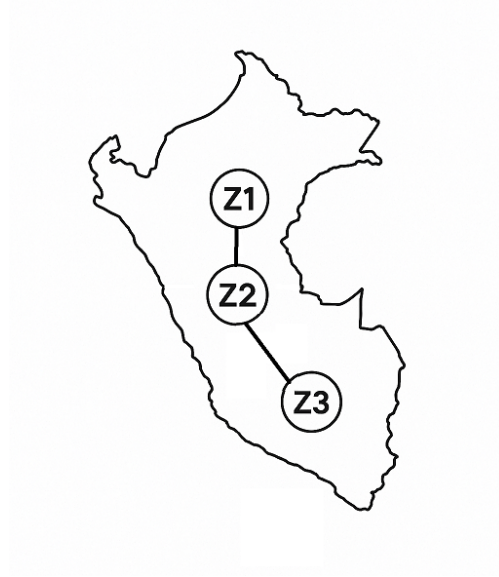


Figure 1: Zonal representation of the Peruvian power system: north (Z1), centre (Z2) and south (Z3) in series. All exchange between the extremes transits the centre.

Table 1: Initial condition, January 2025, as represented in the model. Technology shares are given in Supplementary Material S2.

Zone	Description	Capacity [MW]	Share
Z1	North	1,606.1	11.4 %
Z2	Centre	8,461.9	60.2 %
Z3	South	3,989.0	28.4 %
Total		14,057.0	100 %
<i>Corridors: Z1–Z2 900 MW over 232.3 km; Z2–Z3 700 MW over 271.8 km</i>			
<i>Network: 16,180 km total; thermal and reservoir hydro are 78 % of capacity, wind and solar 10.7 %</i>			

not. The auction mechanism that for a decade channelled renewable investment has been suspended, leaving developers without a reliable off-take signal [8, 45], and the grid was not built with variable renewables in mind. This combination is not unusual: institutional quality and the cost of capital, rather than resource quality, condition deployment [6, 11].

3.2. Zonal structure and transmission

The model represents the system as three zones connected in series (Figure 1). This is not a simplification for tractability but the physical topology: Peru’s main grid runs longitudinally, so the southern zone reaches the rest of the system only through the centre. Table 1 reports the January 2025 starting condition. The centre zone holds 60 % of installed capacity; the Z1–Z2 and Z2–Z3 corridors provide 900 MW and 700 MW of transfer capability; there is no direct Z1–Z3 path. A series topology has no redundancy: a constraint on the central corridor isolates a third of the system, and every transfer between the extremes consumes capacity on both corridors.

Congestion management in Peruvian planning has become more sophisticated, with $n - 1$ contingency screening alongside demand, generation, fuel-price and weather projections [46]. At regional level, Andean

interconnection is technically and economically feasible but has stalled on regulatory sovereignty rather than engineering [37, 19].

3.3. Planning challenges and the modelling implication

The gains from the 1990s reform were real but uneven. Once the initial wave of privatisation ran its course, investment in generation and transmission slowed considerably, indicating that the regulatory framework was not generating the long-term certainty infrastructure capital requires [34]. The suspension of the auction mechanism has hit onshore wind hardest: projects conceived for structured procurement now compete in a spot market under price uncertainty, and the best wind resources remain far from consumption [45]. Fossil dependence and a renewable share of roughly 5.5 % of total energy leave Peru exposed to external price swings [9].

This is the feature that motivates the modelling choice in Section 4. Lead times here are an institutional outcome rather than an engineering parameter, and they differ systematically between generation, which responds to a market signal, and transmission, which is planned centrally and delivered on a longer clock. Representing that asymmetry explicitly is what allows the model to be asked which of the two binds.

Peru is instructive comparatively: simultaneously a reference case for successful market reform and a cautionary example of its limits — competition and improved security of supply, but increasing fossil dependence and a high reserve margin [34]; early leadership in renewable auctions, but no consolidated transition strategy, a gap visible against Chile, which started from a comparable base and has moved further [31, 19].

4. Methodology

4.1. Approach

The question requires two things usually provided separately: a model whose behaviour arises from the structure of the decision process rather than from an optimality assumption, because the phenomenon of interest is the mismatch between a decentralised generation market and a centrally planned network; and a designed experiment over the inputs, because a ranking cannot be read off a handful of named scenarios.

We therefore combine a system dynamics model with a Latin-hypercube designed experiment. The simulation runs monthly from January 2025 to January 2050 (300 months, integration step 0.25 month), with an hourly dispatch represented inside each month by one representative day of 24 hourly levels. The experiment comprises 129 runs: one central case, two one-at-a-time runs per factor, and 120 Monte Carlo samples in which all four factors vary simultaneously. One-at-a-time runs isolate each factor against a fixed reference, which makes a tornado bar attributable but says nothing about interaction; the Monte Carlo ensemble moves everything at once, which the percentile envelopes and rank correlations require but leaves individual effects to be inferred statistically. Reporting both, and checking they agree, is stronger than either alone.

4.2. Model structure

Figure 2 shows the modules, flows and governing parameters. The full equation set is in Supplementary Material S1 and the complete parameter set, with the workbook cell of every value, in Supplementary Material S2.

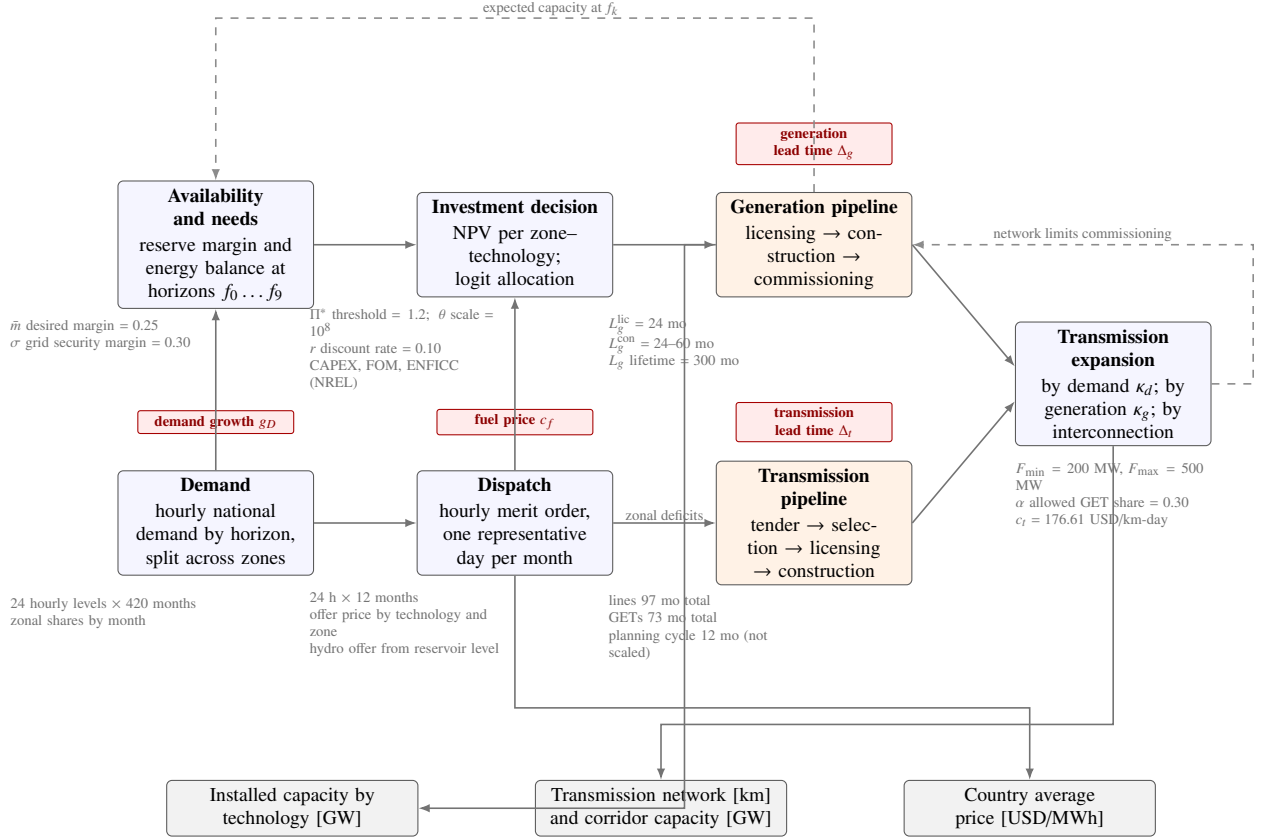


Figure 2: Structure of the model. Solid arrows are material and information flows within a simulation step; dashed arrows are the two feedbacks that generate the behaviour of interest — commissioning is constrained by the network that has actually been delivered, and investment decisions are taken on expected rather than current capacity. Grey text gives the governing parameters and their central values; the four boxes in red are the uncertain drivers varied in the designed experiment of Section 4.3.

4.2.1. Generation: investment, pipeline, commissioning

Generation capacity is a stock fed by a pipeline in which committed projects pass through licensing and construction before commissioning, with total lead time $(L_{z,g}^{lic} + L_{z,g}^{con})(1 + \Delta_g)$ — licensing at 24 months for every technology, construction from 24 months (PV) to 60 (reservoir hydro). Commissioning is bounded by the network actually delivered: capacity that has finished construction cannot enter until there is transmission to evacuate it. This is the first of the two feedbacks drawn dashed in Figure 2, and it is what couples the generation market to the network plan.

Investment is triggered two ways. Under adequacy, a shortfall of the expected reserve margin below the desired margin $\bar{m}$ becomes a capacity requirement. Under profitability, the model computes a net present value per zone–technology over a 25-year life — assuming the plant is the only entrant and that current plus committed capacity persists, with CAPEX, fixed O&M, fuel and firm-capacity parameters from National Renewable Energy Laboratory [27] — admits options above a threshold Π^* , and allocates them across technologies by a logit share with scale θ . That share represents the dispersion of investor beliefs rather than a single optimising agent, and is the point at which the model departs most clearly from a GEP formulation. Decisions use *expected* rather than current state: expected capacity at 3, 5, 7 and 9 years includes the pipeline, so investors know the construction times they face. This second feedback prevents over-investment against a deficit already being remedied.

4.2.2. Transmission: three categories, three triggers

Transmission expansion is disaggregated because the categories respond to different drivers and carry different lead times. Network *for demand* follows zonal peak growth at an empirical intensity κ_d and carries no construction delay, since the model does not represent intra-zonal networks. Network *for generation* connects each new plant at intensity κ_g , triggered by the formal identification of a project at a planning horizon — structurally *reactive*, responding to confirmed rather than anticipated projects, which combined with transmission lead times exceeding generation lead times is a principal source of the bottleneck studied here. Network *for interconnection* is triggered when a transfer implied by the dispatch exceeds delivered corridor capability, met by a grid-enhancing technology if the shortfall is at most a share α of current capability and by a new line otherwise, with increments bounded by $F_{\min}$ and $F_{\max}$ to represent lumpiness.

Transmission projects pass a four-stage pipeline — tender, investor selection, licensing, construction — totalling 97 months for lines and 73 for GETs, each stage scaled by $(1 + \Delta_t)$. The 12-month planning cycle is deliberately not scaled: the model uses it as the period of a modulo operation setting when plans are revised.

4.2.3. Dispatch, prices and distributed resources

Each month the model dispatches 24 hourly levels by merit order, one plant per zone–technology pair; reservoir hydro offers between a minimum and a scarcity price as a function of storage level, giving hydrology a price signal rather than merely an energy constraint. An unconstrained dispatch sizes the transfers the network would ideally carry and hence detects reinforcement needs; a constrained one respecting delivered corridor capability yields congestion, zonal prices and the realised mix. The reported price is the country average tariff over the final twelve months, averaged because it oscillates with hydrology. An external path solving the constrained dispatch with PyPSA [7] exists but is unused here.

Distributed PV adoption is a diffusion process driven by the gap between the levelised cost of distributed generation and the retail tariff, moderated by a potential market and a social-influence term [29, 15, 49]. In every run reported here, system-level storage and grid-enhancing technologies are *disabled* at the switch level, as in the shipped baseline, while distributed resources remain endogenous. This bounds the interpretation: the study quantifies uncertainty around the current policy configuration, not the effect of deploying storage or GETs.

4.3. Experimental design

Four drivers are varied (Table 2), selected because each is uncertain, outside the planner’s direct control on the relevant horizon, and argued in the literature to be a principal risk for a system with Peru’s characteristics.

Two conventions matter. The demand factor is a *growth-path* factor, not a level shift, so 2025 demand is untouched and only the rate of growth varies — the correct construction for an expansion study. Generation lead times are expressed relative to the *workbook base case* rather than the planned time: the Peruvian data set carries an “other delays” fraction of -0.2 , so the factor enters as $(1 + \text{base})(1 + \text{change}) - 1$, giving -0.36 and -0.04 at the ends. This is the convention used for transmission, which makes the two delay bars comparable.

Distributions are uniform and independent; there is no information justifying another shape. The Monte Carlo sample is a Latin hypercube [25] generated from a fixed seed and written to disk before any simulation, so the experiment is specified independently of its execution. The realised design has a maximum absolute correlation between factors of 0.181 and a mean of 0.076; near-orthogonality is what makes the rank correlations in Section 5.2 attributable to single factors.

Table 2: Uncertain factors and ranges. Ranges are ranges of admissibility, not estimated distributions.

Factor	Range	Model quantity
Demand growth g_D	$\pm 20\%$	multiplier on national demand, equal to 1 in January 2025 and $1 + g_D$ in January 2050, so only the growth path changes
Generation lead time Δ_g	$\pm 20\%$	licensing plus construction time of every generation technology
Transmission lead time Δ_t	$\pm 20\%$	tender, investor selection, licensing and construction of lines and GETs
Fuel price c_f	$\pm 15\%$	fuel cost of every technology that burns fuel

4.3.1. Central, best and worst case

The **central case** is the system run with every factor at its reference value, the configuration of the input data set as compiled from Peruvian sources. It is not a forecast and not the ensemble median: it is the reference against which every other run is measured, and the tornado bars are departures from it. The **best** and **worst** cases are selected from the 120 Monte Carlo runs on a single declared metric — the lowest and highest country average price over the final twelve months — which keeps the selection auditable, where a weighted composite would make it depend on weights with no defensible basis. That “best” on price need not be best on every indicator is a result, not a defect (Section 5.4).

4.4. Implementation and validation

The model is implemented in Vensim DSS [48] and executed through its DLL from a Python driver, each uncertain factor entering as a named constant multiplying the corresponding structural equation with a default value that reproduces the unmodified model exactly. Model confidence is established by the standard system dynamics battery [5, 28, 32] rather than a single fit statistic — structure and parameter verification, dimensional consistency, six extreme-condition tests, behaviour reproduction and behaviour sensitivity — reported in full in Supplementary Material S3. Re-simulating the central case on the re-published four-knob model returns 63.85 GW and 46,341.23 km at January 2050, matching the reference exactly.

5. Results

5.1. Central case: the trajectory the reference data imply

Under the reference values of every factor, the Peruvian system roughly quadruples its installed generation capacity between 2025 and 2050, from 14.06 GW to 63.85 GW, while the transmission network grows from 16,180 km to 46,341 km and inter-zonal transfer capability from 1.60 GW to 4.56 GW. The country average electricity price falls from roughly 210 USD MW⁻¹ h in the late 2020s to 124.5 USD MW⁻¹ h by 2050.

The composition of that growth is the first result worth noting. Almost all of it is onshore wind and solar: wind rises from 1.02 GW to 34.83 GW and solar PV from 0.48 GW to 8.33 GW, while thermal capacity grows only from 7.25 GW to 9.65 GW, so non-thermal technologies hold 84.9 % of capacity by 2050 against 48.4 % at the start. The model reaches this without any renewable target, mandate or carbon price: expansion is driven by the profitability rule alone. Thermal plant is not retired — the model has no endogenous retirement flow — but it stops being the marginal investment.

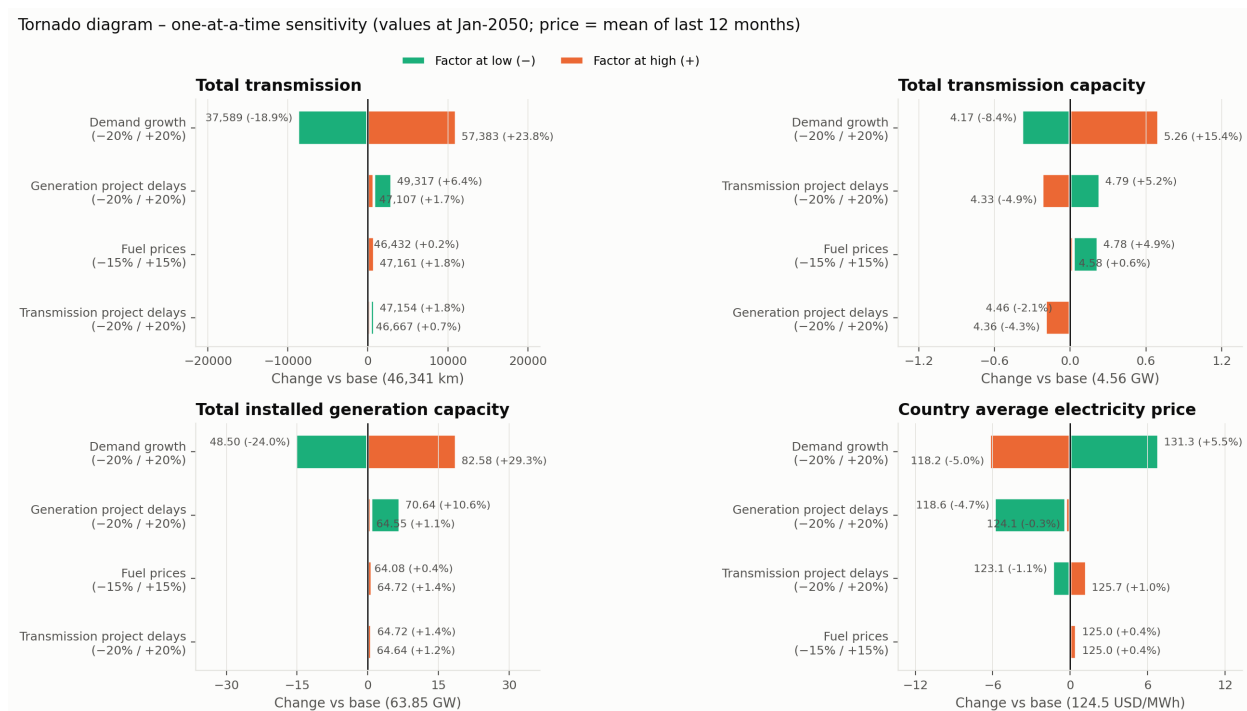


Figure 3: Tornado diagram. For each indicator, the bars show the change from the central case when one factor is set to the low and the high end of its range with the other three held at their reference value. Factors are ordered by swing, $|\text{high} - \text{low}|$. Values at January 2050; the price is the mean of the final twelve months.

The price path is not monotone: it rises through the late 2020s, when demand grows against a capacity base that cannot yet respond, and falls thereafter as wind and solar enter. That is the classic capacity-cycle signature system dynamics models of liberalised markets reproduce [16, 2, 40], and it appears here without being imposed.

5.2. Which uncertainty governs the system

Figure 3 reports the one-at-a-time sensitivity of each indicator. The result is unambiguous: demand growth dominates by roughly an order of magnitude over the next-ranked factor, and fuel prices — the risk most often emphasised for a gas-dependent system — are the weakest driver of three of the four indicators.

Table 3 gives the numbers. A $\pm 20\%$ band on the 2050 demand level moves required transmission from -18.9% to $+23.8\%$ and installed capacity from -24.0% to $+29.3\%$; the swing attributable to demand is nine times that of generation lead times for transmission length. Fuel prices move the average price by 0.4% in both directions — not a sign error but a genuine non-monotonicity.

The result is confirmed when all four factors move together (Table 4): demand growth reaches 0.985 for transmission length and 0.973 for installed capacity, close to the maximum a rank correlation can attain, while no other factor exceeds 0.27 in absolute value. The two methods agree, which is the check that matters — a one-at-a-time result surviving only in the absence of interaction would not support the conclusion.

5.3. How wide the range of outcomes remains

Figure 4 shows the P10–P90 envelope of each indicator. The envelopes are narrow until the early 2030s and widen steadily thereafter — the expected signature of a system whose divergence is driven by a growth rate rather than a level.

Table 3: One-at-a-time sensitivity. Change from the central case at January 2050 with one factor at each end of its range. Factors ordered by swing within each indicator.

Indicator	Factor	Low	High	Swing
Transmission km, base 46,341	Demand growth	−18.9%	+23.8%	19,794
	Generation lead time	+6.4%	+1.7%	2,210
	Fuel price	+0.2%	+1.8%	729
	Transmission lead time	+1.8%	+0.7%	487
Installed capacity GW, base 63.85	Demand growth	−24.0%	+29.3%	34.08
	Generation lead time	+10.6%	+1.1%	6.09
	Fuel price	+0.4%	+1.4%	0.64
	Transmission lead time	+1.4%	+1.2%	0.08
Transfer capability GW, base 4.56	Demand growth	−8.4%	+15.4%	1.08
	Transmission lead time	+5.2%	−4.9%	0.46
	Fuel price	+4.9%	+0.6%	0.19
	Generation lead time	−2.1%	−4.3%	0.10
Average price USD MW ^{−1} h, base 124.5	Demand growth	+5.5%	−5.0%	13.09
	Generation lead time	−4.7%	−0.3%	5.52
	Transmission lead time	−1.1%	+1.0%	2.63
	Fuel price	+0.4%	+0.4%	0.01

Table 4: Spearman rank correlation between each factor and each indicator over the 120 Monte Carlo runs, in which all four factors vary simultaneously.

Factor	Transmission [km]	Transfer [GW]	Capacity [GW]	Price [USD MW ^{−1} h]
Demand growth	0.985	0.835	0.973	−0.752
Transmission lead time	0.112	−0.142	0.121	−0.081
Generation lead time	0.044	0.081	0.053	−0.264
Fuel price	−0.008	−0.013	−0.014	−0.014

By 2050 the ranges are substantial (Table 5): installed capacity spans 52.3 GW to 82.5 GW between the tenth and ninetieth percentile, a factor of 1.6, and required transmission 39,858 km to 57,069 km. The price range is comparatively narrow, 116.8 USD MW^{−1} h to 129.4 USD MW^{−1} h. That asymmetry is itself informative: what these uncertainties move is *how much infrastructure the country has to build*, far more than *what electricity ends up costing*.

The central case sits close to the median on transmission and capacity but at the upper end of the price distribution. That is a consequence of the price non-monotonicity identified in Table 3: because both an increase and a decrease in several factors lower the price relative to the reference, the reference is near a local maximum.

Supplementary Material S4.2 reports the convergence of these percentiles as runs are added. P10, P50

Table 5: Indicator values at January 2050 over the 120 Monte Carlo runs.

Indicator	Central	P10	P50	P90	P90−P10
Transmission [km]	46,341	39,858	47,788	57,069	17,211
Transfer capability [GW]	4.56	4.23	4.70	5.29	1.06
Installed capacity [GW]	63.85	52.33	66.82	82.49	30.16
Average price [USD MW ^{−1} h]	124.5	116.8	122.8	129.4	12.59

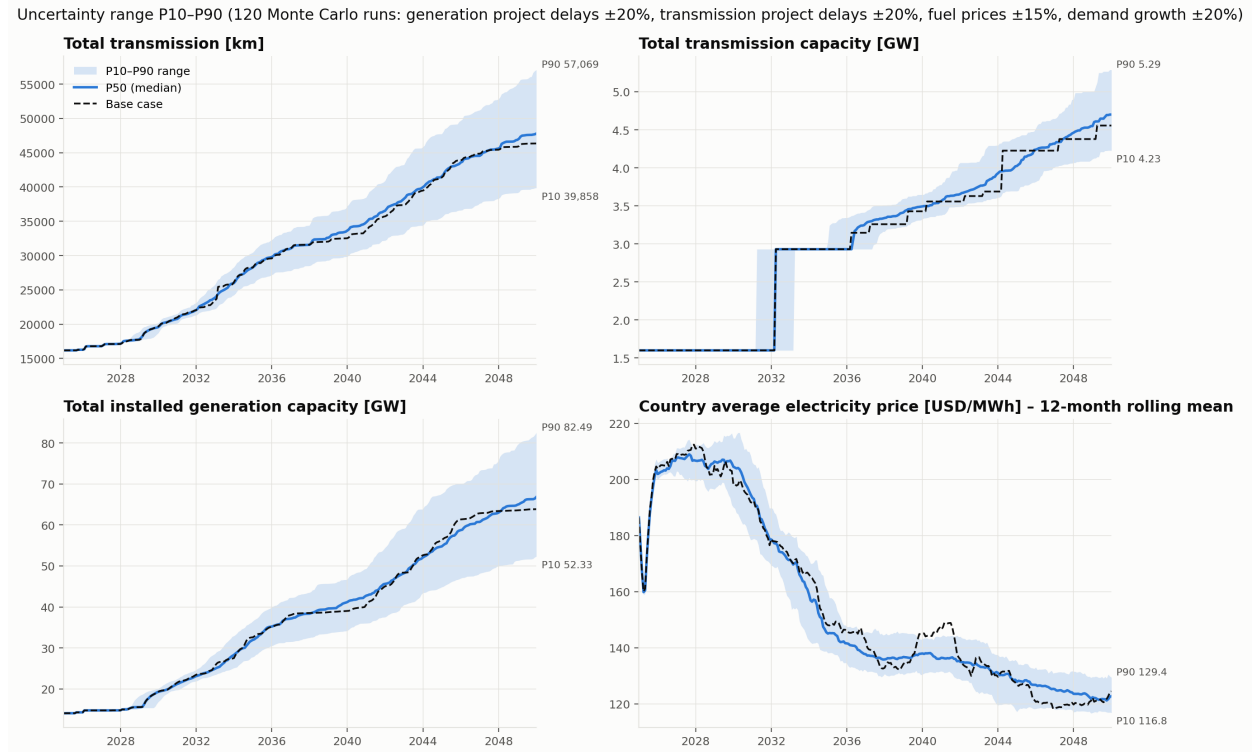


Figure 4: P10–P90 envelope of the four indicators over 120 Monte Carlo runs, with the ensemble median and the central case. The band contains the central 80 % of outcomes.

and P90 are stable from roughly 75 runs onwards for every indicator, which is the evidence that 120 samples are sufficient rather than an assumption that they are.

5.4. Central, best and worst case

Table 6 reports the three trajectories selected as described in Section 4.3.1. The best case on delivered price is run mc_009, in which demand grows 17.3 % faster than the reference; the worst is mc_058, in which it grows 16.5 % slower.

This is the central empirical finding, and it runs against the intuition that higher demand means a more stressed and therefore more expensive system. The mechanism is visible in the technology mix (Figures 5 and 6): in the best case onshore wind reaches 52.05 GW against 34.83 GW centrally and 25.47 GW in the worst, and solar PV 11.51 GW against 8.33 and 5.24 GW. Faster demand growth sustains a price signal long enough for low-cost renewable capacity to clear the profitability threshold earlier and in greater volume, and once in place that capacity depresses the marginal price. Slow demand growth suppresses the investment signal and leaves the system leaning on thermal plant — 10.2 TW h yr⁻¹ in the worst case against 5.6 TW h yr⁻¹ in the best — ending with a higher average price despite a smaller system.

The corollary is uncomfortable for a planner. The least-cost trajectory is also the most infrastructure-intensive: the best case requires 58,562 km of network, 26.4 % more than the central case and 50 % more than the worst case, together with 10.3 % more inter-zonal transfer capability. The cheapest electricity in this system is not obtained by building less; it is obtained by building more, faster. Figure 7 shows the resulting price paths and Figure 8 the transmission requirement.

Table 6: Central, best and worst case. Best and worst are the Monte Carlo runs with the lowest and the highest country average price over the final twelve months. Percentages are relative to the central case.

	Central	Best (mc_009)	Worst (mc_058)
<i>Factor values</i>			
Demand growth	0.0%	+17.3%	-16.5%
Generation lead time	0.0%	+9.8%	-5.4%
Transmission lead time	0.0%	+2.6%	+8.2%
Fuel price	0.0%	-2.5%	-1.1%
<i>Indicators at January 2050</i>			
Transmission [km]	46,341	58,562 (+26.4%)	38,927 (-16.0%)
Transfer capability [GW]	4.56	5.03 (+10.3%)	4.34 (-4.8%)
Installed capacity [GW]	63.85	86.40 (+35.3%)	49.18 (-23.0%)
Average price [USD MW ⁻¹ h]	124.5	114.9 (-7.6%)	135.6 (+9.0%)

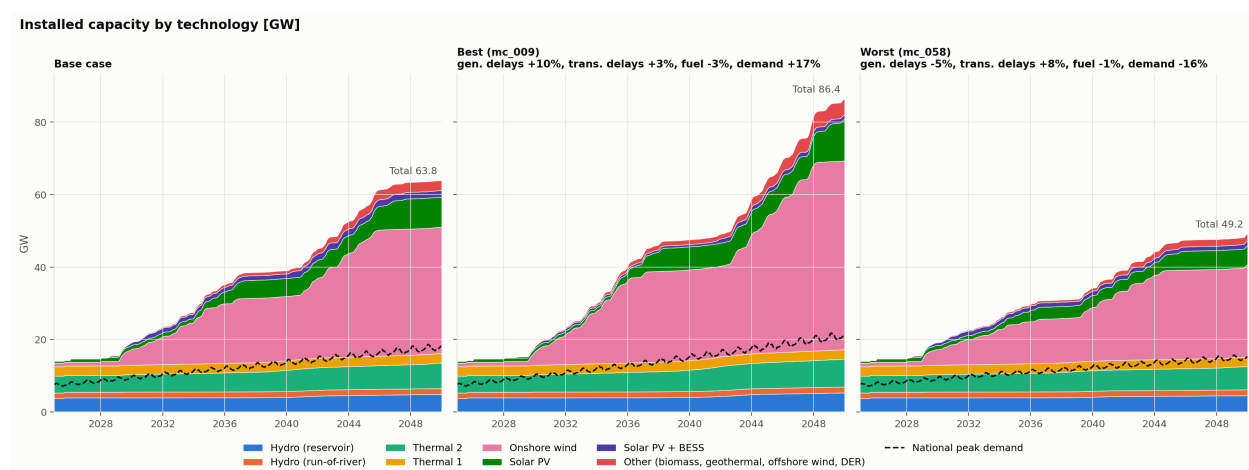


Figure 5: Installed capacity by technology for the central, best and worst case. The dashed line is national peak demand. Fourteen model technologies are aggregated into eight series; the mapping is given in Supplementary Material S5.3.

A second observation concerns the delay factors. The best case has *longer* generation lead times than the central case (+9.8 %) and the worst case has shorter ones (-5.4 %). This is not evidence that delays are beneficial; it is a consequence of their weakness relative to demand. Within the sampled ranges, the demand draw determines the outcome and the delay draw is close to noise, so the extreme runs on price carry essentially arbitrary delay values. Reporting this explicitly is more honest than selecting extreme runs that happen to tell a tidier story, and it is itself the clearest statement of the ranking: in this system, over these ranges, lead times do not decide the outcome.

5.5. Where the uncertainty acts

Decomposing transmission by category (Figure 8) locates the effect. Network built for demand connection scales directly with peak demand and inherits the demand factor's variance in full; network built to connect generation follows the investment decision and inherits it indirectly. Inter-zonal corridor capability is the least sensitive of the three and the only one where transmission lead time ranks second rather than fourth (Table 3): a longer pipeline defers reinforcements, which the series topology converts into persistent congestion rather than a different final capacity. That is why transmission lead time barely registers in the

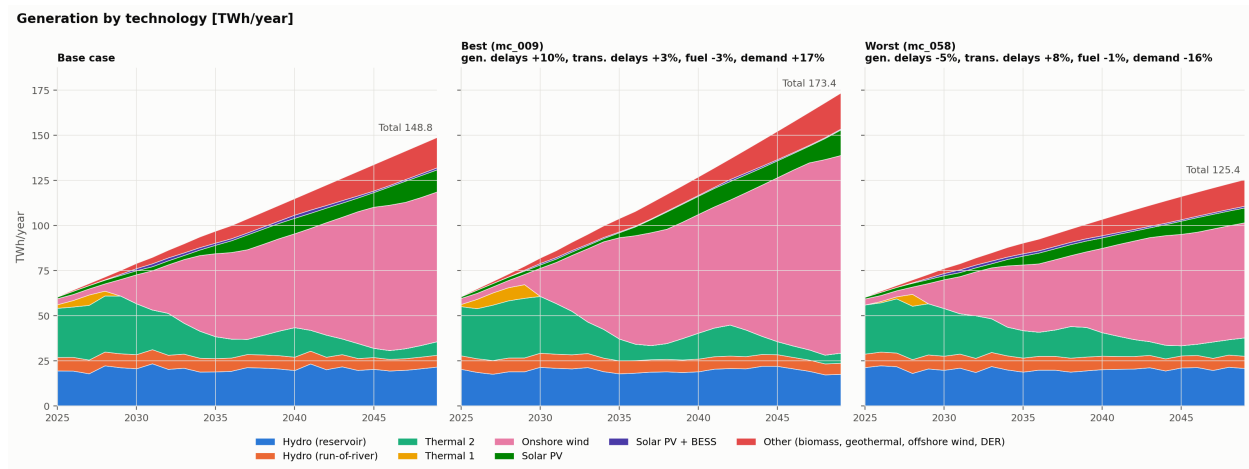


Figure 6: Annual generation by technology for the central, best and worst case.

aggregate indicators while remaining operationally important — the model builds much the same network in the end, and what the lead time changes is *when* it is available.

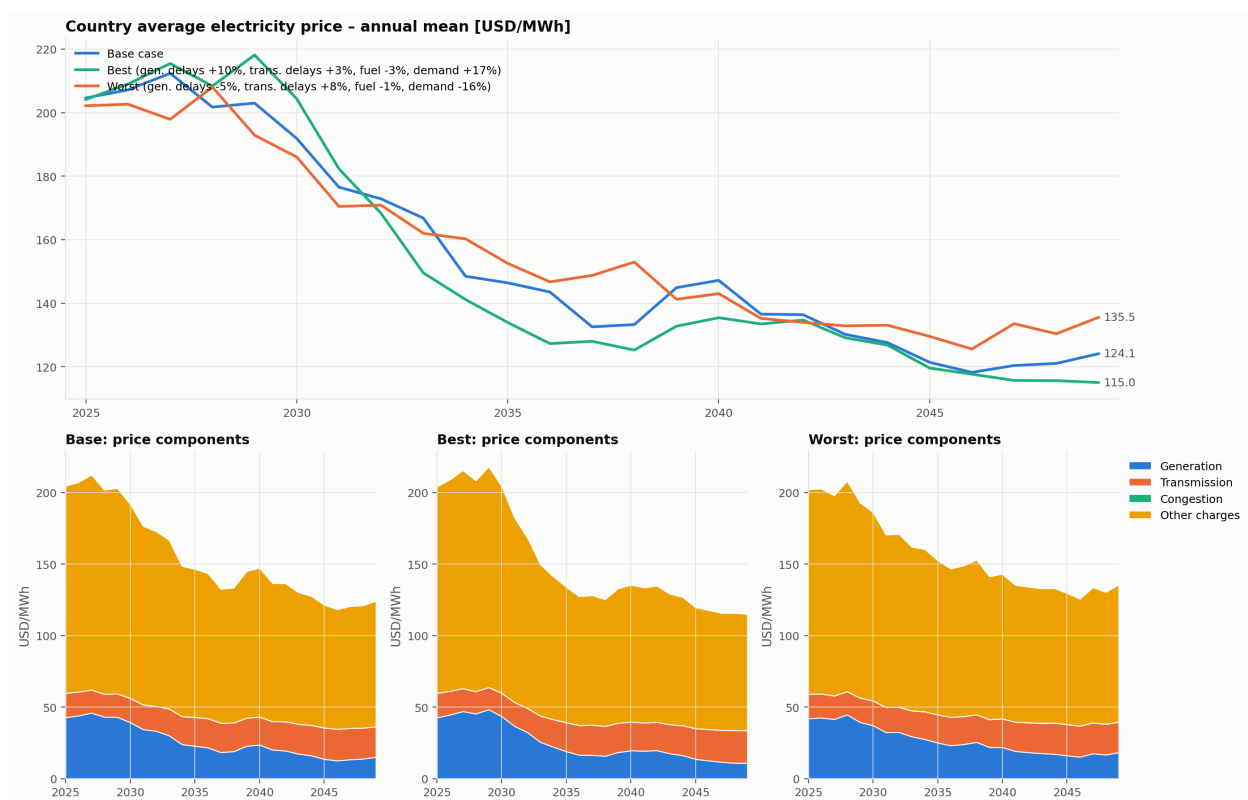


Figure 7: Country average electricity price for the central, best and worst case.

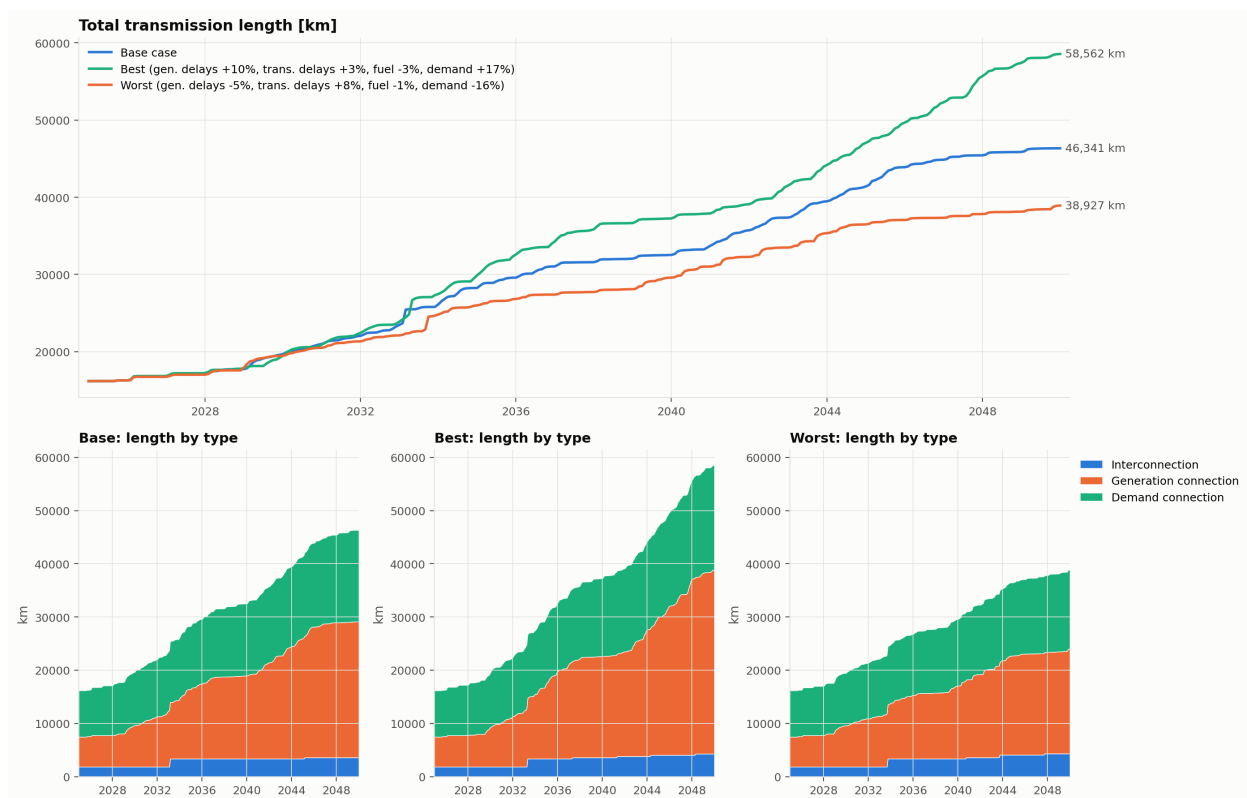


Figure 8: Transmission network by category — demand connection, generation connection and interconnection — for the central, best and worst case.

6. Discussion

6.1. Demand growth as the governing uncertainty

That demand dominates is structurally explicable. It enters at the root of every chain: it sets the adequacy trigger, the residual demand the dispatch clears and hence the price signal driving the profitability trigger, and the zonal peaks that drive demand-connection network. Fuel prices enter only at the dispatch, and in a system that ends 85 % non-thermal a cost shift on the shrinking thermal margin has little leverage — the rank correlation of -0.014 is what one should expect once the marginal technology is no longer the one burning the fuel.

The reading is uncomfortable: demand growth is the one driver planning does not control and forecasting handles poorly, and our ± 20 % range is arguably conservative for an emerging economy with incomplete electrification and potential industrial loads. Long-horizon demand uncertainty [1] is therefore the main lever on the quality of the answer, not a peripheral refinement.

6.2. The price result and what it does not say

The finding needs careful statement. It concerns a long-run *average* under an investment rule responding to profitability: faster demand sustains the price signal long enough for low-cost capacity to clear the threshold earlier and in greater volume, and that capacity then depresses the marginal price. It does not say demand growth is costless — the same runs require 26 % more transmission and 35 % more generation capacity, and the model prices neither the capital at risk nor the land and social cost of that build-out — nor anything about short-run prices: Figure 7 shows the high-demand case *above* the others through the late 2020s.

Three structural caveats bound it. The model has no endogenous retirement, so the mix shifts by addition rather than replacement. Storage and grid-enhancing technologies are disabled in every run, so the substitution between flexibility and network documented by Sousa et al. [41] and Sadeghian and Mohammadi-Ivatloo [35] is absent; enabling them would plausibly reduce the transmission requirement of the high-demand case and weaken exactly the corollary we emphasise. And the dispatch uses one representative day per month, which under-represents sustained scarcity.

6.3. Delays, and what the indicators miss

Lead times rank second and third, well behind demand. We do not read this as evidence that delays are unimportant for Peru.

What the result says is that, over a ± 20 % band around current practice, lead times do not change the *size* of the system built by 2050 — not that they leave its condition unchanged along the way. Section 5.5 shows the mechanism: longer pipelines defer reinforcements, and in a series topology the deferral surfaces as inter-zonal congestion rather than a different terminal capacity. An indicator measured at January 2050 is poorly suited to capture that, which is a limitation of our reporting as much as a property of the system; a study aimed at reliability would need congestion-hours, curtailment and unserved energy.

There is also a range effect. A ± 20 % band is plausible for incremental administrative improvement but not for institutional failure: the literature on delay causes [43, 6] describes multi-year slippage on individual projects, well outside ± 20 % of a planned stage. Our ranking is conditional on the ranges in Table 2, as any sensitivity ranking must be.

6.4. Fuel-price exposure reconsidered

Peru's gas dependence is the feature most often identified as its principal vulnerability [34, 13, 9], and our results qualify that view temporally rather than substantively. Fuel prices are the weakest driver on every indicator because the model's own trajectory removes the exposure: by 2050 thermal plant supplies 7.5 TW h yr^{-1} of $148.8 \text{ TW h yr}^{-1}$. Fuel-price risk is therefore a near-term exposure that the expansion path itself retires — provided the path is delivered. If the network cannot be built at the rate the investment rule assumes, the system stays thermal for longer and the exposure persists. Read that way, fuel-price risk in Peru is a *consequence* of transmission delivery rather than an independent risk to hedge separately, which is a different prescription from the usual one.

6.5. Renewable integration and coordination

The central case reaches 84.9 % non-thermal capacity by 2050 with no renewable target, driven by profitability alone — a stronger statement about the competitiveness of Peruvian wind and solar than a target-constrained model could make, and one that aligns with the optimisation-based findings of Fiestas-Chevez et al. [14] and De la Cruz Torres et al. [10] from a different methodological direction. Where the two diverge is on what getting there costs: an optimisation model coordinates generation siting and network expansion by construction, while ours does not, because the Peruvian institutional arrangement does not. Herding et al. [20] quantify the prize for closing that gap — around 10 % lower investment and up to 30 % less network expansion — and Sauma and Oren [36] formalise the proactive-versus-reactive distinction underlying it. Peru is firmly on the reactive side, and the cost of remaining there rises with demand growth. Complementary options attack the constraint from the demand side: distributed generation reduces the transfers a longitudinal grid must support [15, 49, 38], and grid-enhancing technologies raise corridor capability without new right-of-way [35]. Both are represented but disabled here; quantifying them is the obvious next study, and our results identify where the gain would be largest — the Z2–Z3 corridor under high demand.

6.6. Policy implications and comparison

Four implications follow. *Transmission delivery capacity is the planning variable to protect*: demand governs the outcome and is outside the planner's control, while the least-cost outcome is the most network-intensive, so under-building the grid forecloses the cheap trajectory. *Anticipatory network planning has measurable value*, since the reactive trigger plus the lead-time asymmetry is a structural bottleneck rather than a parameter choice [36, 20]. *Demand forecasting deserves the effort currently given to fuel-price hedging*, given a ratio of roughly 1:900 in rank correlation between the two. And *restoring a predictable procurement signal matters more than the price it clears*, since the profitability trigger responds to a sustained signal that the suspended auction mechanism removes [8, 45, 11].

Peru's series topology drives the result and is unusual regionally: Brazil's meshed system offers alternatives when a corridor binds, so the constraint appears as regional heterogeneity [39] or curtailment [21] rather than a single critical path, while Chile — also longitudinal — is the more instructive comparison and has moved further from a similar base [31]. The common regional thread is institutional [19, 37, 6], and our finding should not be generalised to systems with weaker institutions or wider delay distributions than we sampled.

6.7. Methodological implications

Pairing a system dynamics model with a designed experiment changes what the model can claim: named scenarios show what a model does but cannot separate a structural result from an artefact of one input assumption. Here it turned a plausible narrative — that a gas-dependent system faces fuel-price risk — into

a measured ranking that contradicts it, for 129 mechanisable runs with the design fixed before execution. Reporting both one-at-a-time and all-factors-varying evidence is worth the redundancy, since their agreement is what makes the ranking defensible. The execution infrastructure — resumable runs, an archived design, explicit checks that each run produced data — was likewise not incidental: those checks caught silent failures that would otherwise have entered the results as missing data rather than as errors.

7. Conclusions and policy implications

We set out to establish which uncertainty governs the Peruvian power system, not to forecast it. A system dynamics model with explicit generation and transmission pipelines, subjected to a 129-run designed experiment over 2025–2050, answers unambiguously.

Demand growth dominates, by an order of magnitude. It reaches Spearman rank correlations of 0.985 with required transmission, 0.973 with installed capacity and -0.752 with the average price, while no other factor exceeds 0.27 in absolute value. A $\pm 20\%$ band on the 2050 demand level moves required transmission by -18.9% to $+23.8\%$ and installed capacity by -24.0% to $+29.3\%$. The one-at-a-time and all-factors-varying evidence agree on this ranking.

Fuel-price risk is weaker than the literature on Peru implies, for a structural reason. Fuel prices are the weakest driver on three of four indicators, moving the 2050 average price by $0.005 \text{ USD MW}^{-1} \text{ h}$, because the expansion path itself removes the exposure: thermal generation falls to 7.5 TW h yr^{-1} of $148.8 \text{ TW h yr}^{-1}$ by 2050. Fuel-price risk is a near-term exposure contingent on transmission delivery rather than an independent long-run risk.

The least-cost trajectory is the most network-intensive. Faster demand growth lowers the long-run average price by sustaining the price signal long enough to pull forward low-cost wind and solar. The best case on delivered price requires 58,562 km of transmission, 26.4 % more than the central case and 50 % more than the worst, with 10.3 % more transfer capability. For a system whose grid is already binding, the cheap path is the one that demands most of the planner.

Lead times change when, not how much. Transmission lead times rank last or second-to-last on terminal indicators because the model builds much the same network either way; the deferral surfaces as inter-zonal congestion rather than a different system size. Terminal stocks are the wrong indicator for that effect.

The expansion is renewable without being told to be. The central case reaches 84.9 % non-thermal capacity by 2050 under a profitability rule alone, with no target, mandate or carbon price, agreeing in direction with the optimisation-based Peruvian literature [14, 10].

Peruvian planning attention is therefore not proportionate to the sensitivities. Effort on fuel-price exposure buys little; effort on long-horizon demand forecasting, on the institutional capacity to deliver transmission at the rate demand requires, and on moving the network trigger from confirmed to anticipated projects [36, 20] buys a great deal.

The ranking is conditional on the sampled ranges. Storage and grid-enhancing technologies are disabled in every run, so the flexibility–network substitution is absent and the transmission requirement of the high-demand case is, if anything, overstated; retirements are exogenous, the dispatch uses one representative day per month, and the distributions are ranges of admissibility rather than estimated densities. Each is documented with its consequence in the supplementary material. Three extensions follow: enabling storage and grid-enhancing technologies under the same design; replacing terminal stocks with congestion-hours, curtailment and unserved energy to capture the timing effect of lead times; and applying the design to Brazil and Chile with the same model structure, to separate what is Peruvian from what is a property of longitudinal systems generally.

Data and code availability

The Vensim model, the complete input workbook, the experimental design, the simulation driver, the post-processing code and the commented notebook that regenerates every figure in this paper are openly available at <https://github.com/sebastianzapatar/peru-expansion-uncertainty>. **[Replace with the DOI of the archived release before submission.]** Nothing required to reproduce the results is available only on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to draft and edit sections of the manuscript, to structure the experimental design and analysis code, and to prepare figures and tables from the simulation output. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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